

Rahma App Technical Clarification Notes

Offline mode, AI model usage, compliance discovery, intent detection, public services, and open implementation questions based on the Rahma proposal and current project discussion. Date: 21 May 2026.

Core product idea	One-touch emergency assistance for elderly users, children, and people with special or chronic needs.
Architecture claim to clarify	Hybrid on-device/cloud AI, offline emergency capability, SMS fallback, UAE-hosted health data, and secure integrations with healthcare and emergency systems.
Main purpose of this note	Identify what must be clarified before implementation so Phase 1 does not start on unsafe or unrealistic assumptions.

Key position: offline mode should not mean the full Rahma product works without internet. It should mean the emergency-critical path degrades safely when internet, GPS, speech recognition, or AI services are unavailable.

1. Offline Working: What Should Work Without Internet

The proposal says the core emergency function operates with zero internet via SMS fallback. This needs a precise definition. From an implementation perspective, offline mode should be split into must-work, degraded, and not-supported categories.

Feature area	Expected offline behavior	Implementation note
Emergency activation button	Must work offline. User can press and hold the button even without data connectivity.	The app should create a local emergency event and try SMS/call fallback immediately.
Local emergency profile	Must work offline for minimum fields.	Store limited encrypted data on device: name/ID reference, language, medical flags, allergies, caregiver numbers, and emergency contacts.
GPS/location	Degraded offline.	Use last known location, GPS if available, network location if available, timestamp, and location confidence. If no location is available, send fallback message without location plus last known area.
Caregiver alert	Degraded offline.	Use SMS fallback to caregiver numbers. Push notification requires internet and should be retried when connection returns.
Emergency dispatch	Degraded offline.	If direct API needs internet, fallback to emergency call/SMS/call-center handoff. Do not depend only on cloud API.
Voice analysis	Limited offline.	Use simple local prompts and possibly phrase spotting. Avoid relying on offline dialect intent classification for safety-critical decisions.
Hospital finder	Limited offline.	Cache a small approved list of nearby/region hospitals and emergency numbers. Real-time availability is not offline-safe.
AI health triage	Not recommended offline for Phase 1.	Do not run complex medical triage offline unless clinically validated. Use safe scripted instructions only.
Event audit log	Must work offline locally.	Queue encrypted event logs locally and sync when internet returns.

Offline mode recommended flow

- User presses and holds the Rahma button for two seconds.
- App creates local emergency event and captures current/last known location with timestamp and accuracy.
- App reads minimal encrypted local profile and caregiver numbers.
- If internet is unavailable, app sends SMS fallback to caregiver and/or approved emergency handoff number.
- If possible, app opens an emergency call prompt or auto-call flow according to local platform rules and approved policy.
- The app shows simple Arabic/English offline prompts such as Stay where you are, help is being contacted.
- When internet returns, the app syncs the event timeline, alerts, delivery status, and diagnostics to the backend.

Important concern: offline native dialect voice analysis

Offline voice analysis in native dialects/languages may reduce accuracy, especially with elderly speech, stress, background noise, or mixed languages. For Phase 1, offline AI should not be the primary safety decision-maker. The safer design is: button activation always triggers the emergency path, voice only adds context when confidence is high, and uncertainty triggers safe escalation rather than complex questioning.

2. Public AI APIs vs Self-Hosted or Fine-Tuned Models

This cannot be decided only by engineering preference. It depends on client policy, UAE data hosting requirements, health-data rules, the exact data sent to the model, vendor contract terms, and whether the model provider uses data for training or stores it outside approved boundaries.

Option	When it may be acceptable	Risk / condition
Public AI API	Possibly acceptable for non-sensitive development, synthetic test data, translation prototypes, or if the client approves the vendor contract, region, retention, and no-training terms.	High caution. Do not send PHI, location, Emirates ID, voice, transcripts, caregiver data, or emergency events unless formally approved.
Private cloud AI deployment	Better for production if deployed in an approved UAE region or client-controlled environment with enterprise controls.	Still requires vendor review, data residency confirmation, logging policy, and incident response agreement.
Self-hosted model	Best control for sensitive production flows where data must remain in the client platform.	Requires GPU/infra, MLOps, tuning, evaluation, security patching, and model governance.
Hybrid approach	Recommended: on-device/simple offline logic for fallback, private hosted AI for production, public APIs only for approved low-risk use.	Needs clear routing rules so sensitive data does not accidentally leave approved boundaries.

Recommended default for Phase 1: assume production AI must be private, client-approved, and UAE-hosted or explicitly approved by the client. Public AI APIs should be blocked for production PHI unless legal/security signs off.

3. Compliance Documentation Needed Before Starting

Before development starts, the project should create a compliance discovery pack. This is needed because Rahma will process personal data, health data, live location, caregiver access, voice/transcripts, and emergency events.

Compliance area	What to clarify before build
UAE Personal Data Protection Law	Lawful basis/consent, purpose limitation, privacy notice, data subject rights, processing records, cross-border transfer restrictions, and processor/vendor duties.
UAE ICT in Health Fields Law	Health data confidentiality, integrity, availability, authorized access, sharing restrictions, and health-sector data hosting/transfer expectations.
UAE Information Assurance Standard	Security controls, risk assessment, access control, audit, network security, incident response, vulnerability management, and evidence requirements.
Dubai / Abu Dhabi health authority rules	DHA, DoH, Malaffi, NABIDH or other applicable integration policies, depending on exact pilot partners.
AI governance and clinical safety	Model scope, prompt approvals, no-diagnosis boundaries, clinical review process, evaluation test sets, red-team testing, and incident handling.
Child data and school data for Phase 2	Guardian consent, school authority, child location tracking, data retention, and who can view child data.

Official reference anchors to review: UAE Government data protection laws page, UAE Government ICT in health sector page, and UAE Cyber Security Council UAE Information Assurance Standard page. Additional client-specific policies may be stricter than public laws.

4. AI Intent Detection: Robustness and Voting System

A multi-model voting system can improve reliability, but it should not be described as fool proof. No AI intent system is fool proof in emergency conditions. The safer goal is fail-safe behavior: if confidence is low or models disagree, escalate to the safest emergency path.

Layer	Recommended design
Input normalization	Capture audio quality, language hint, profile language, and whether the user is silent or distressed.
Parallel inference	Run multiple intent detectors when online: speech-to-text intent model, direct audio classifier, rules/keyword layer, and profile/context rules.
Voting / consensus	Use weighted voting, not equal voting. Emergency-risk model and rule-based safety layer should override uncertain low-risk outputs.
Confidence policy	If confidence is high, add intent context. If confidence is medium, ask one simple confirmation. If confidence is low or models disagree, escalate safely.
Safety override	Button activation, fall detection, silence after activation, or high-risk medical profile should trigger emergency/caregiver flow even without AI confirmation.
Auditability	Store model versions, confidence scores, final decision, fallback reason, and human override status. Avoid storing raw audio unless approved.
Testing	Evaluate Arabic, English, Gulf dialect, elderly speech, noisy environments, silence, panic speech, and mixed-language utterances.

Recommended Phase 1 rule: AI should enrich emergency context, not decide whether the emergency is real. The one-touch activation itself should be enough to start the safety workflow.

5. Public Services and Auth Options

For a health/emergency app, public social login should be treated carefully. Social auth such as Google, Apple, or Facebook may be convenient, but it may not be appropriate as the primary identity method for patients, vulnerable users, caregivers, or government-linked health workflows.

Auth / public service	Recommendation
UAE PASS	Strong candidate if available and approved. Useful for verified national digital identity and government-aligned workflows. Confirm integration access and user suitability.
Client/government SSO	Recommended for administrators, support users, clinical reviewers, and authority staff. Use the client identity provider where possible.
Mobile OTP	Useful for caregivers and pilot onboarding, but should be combined with caregiver verification and role approval.
Apple/Google social login	Possible only for low-risk convenience flows if approved. Not recommended as the only identity method for medical/emergency access.
Facebook/social auth	Generally not recommended for this use case because it adds privacy and trust concerns without strong identity assurance.
Healthcare identifiers	May be needed for patient matching, but should be handled only through approved health authority workflows and minimum necessary data.
Maps/SMS/push services	Can be used only after data residency, logging, vendor, and privacy review. Sensitive payloads should be minimized.

6. What Looks Good

- The target users are clear and meaningful: elderly users, children, and people with special/chronic needs.
- The one-touch design principle is strong because it reduces cognitive load in emergencies.
- The proposal correctly recognizes multilingual support as critical in the UAE.

- The phased roadmap is useful because it separates pilot launch, child/multilingual expansion, UAE-wide rollout, and GCC expansion.

7. What Needs More Detail

- Exact emergency integration model: direct 998/999 API, government API gateway, call-center handoff, or certified simulator during pilot.
- Legal and clinical responsibility for AI instructions, especially first-aid guidance or health triage.
- Data model and consent model for vulnerable users, caregivers, guardians, schools, and healthcare providers.
- How much voice data is stored, whether transcripts are retained, and how long emergency event data is kept.
- Mobile behavior when the app is closed, phone is locked, internet is unavailable, GPS is delayed, or permissions are disabled.
- Operational support model for false alarms, failed notifications, caregiver questions, privacy requests, and security incidents.

8. Questions to Confirm With Stakeholders

- What exactly must work offline in Phase 1, and what is allowed to degrade?
- Is SMS fallback approved for emergency payloads, and what information can be included in SMS?
- Are public AI APIs allowed for production? If yes, under what data residency, retention, and no-training terms?
- Is self-hosting required for AI models that process voice, transcripts, health data, or location?
- Which laws, client policies, and authority-specific standards are mandatory before development starts?
- Which identity provider is preferred: UAE PASS, client SSO, mobile OTP, or another approved model?
- Who owns clinical approval of emergency wording and AI fallback behavior?
- Which Phase 2 services are highest priority: children/schools, geofencing, pharmacy, reminders, or language expansion?

Sources checked for compliance anchors: UAE Government Data Protection Laws page, UAE Government ICT in Health Sector page, and UAE Cyber Security Council UAE Information Assurance Standard page. This note is for technical planning and does not replace legal or compliance advice.